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# Physics-Informed Sparse Variational Gaussian Processes for Passive Starlink Satellite Detection

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## Abstract

Passive detection of Starlink satellites from ground-based software-defined radio (SDR) captures is a challenging spectrum-monitoring problem: the signal appears as a transient 750 Hz spectral fingerprint in broadband Ku-band noise, and detection confidence varies with orbital geometry and time of day. Classical z-score thresholding (the approach built in a prior course, ECE 257B) achieves near-perfect ranking but is poorly calibrated and gives no uncertainty estimates on ambiguous windows. This work replaces the hard threshold with a physics-informed Sparse Variational Gaussian Process (SVGP) that uses an additive kernel separating spectral, geometric, and temporal evidence. Across 619 705 windows from a 14-day SDR campaign, the GP reduces Expected Calibration Error (ECE) by roughly  $230\times$  compared to the threshold detector ( $9.99\times 10^{-5} \pm 4.95\times 10^{-6}$  vs.  $2.3\times 10^{-2}$ , mean  $\pm$  std over 5 seeds) while retaining perfect AUROC. An ablation study confirms the additive kernel structure is the key design decision: a single RBF over all features is  $5.7\times$  worse in ECE despite identical ranking. Pass-level aggregation of GP probabilities yields zero false positives at  $F1 = 0.952$ , and GP probabilities on held-out SOFT windows correlate strongly with whether the underlying satellite pass was independently confirmed positive.

## 1 Introduction

Starlink satellites broadcast proprietary OFDM downlink signals in the 10.7–12.7 GHz Ku band. The frame structure repeats at 750 Hz, creating a detectable spectral peak in passive SDR captures even when the signal is not decodable. Prior work (ECE 257B, 2025) built a complete passive detection pipeline: a USRP X310 records 50 MS/s baseband, a Welch periodogram accumulates power spectral density (PSD), and a z-score test flags windows where the 750 Hz bin is anomalously elevated relative to its local spectral neighborhood [1].

That pipeline has two practical weaknesses. First, the hard threshold produces a binary output with no probability or uncertainty; it cannot communicate *how confident* it is. Second, roughly 70 000 ambiguous windows (z between 2.5 and 4.0) are entirely discarded: the threshold cannot say whether they are real detections or noise. For a track-before-detect system—which integrates weak evidence over multiple consecutive windows to confirm a satellite overpass before firing an alert—these borderline windows are precisely the most valuable ones, and the threshold discards them entirely.

This project, developed for ECE 228 (Machine Learning for Physical Applications), replaces the threshold with a Gaussian Process classifier that: (1) provides calibrated detection probabilities, (2) encodes physical priors through an additive kernel decomposition, (3) scales to  $\sim 40\,000$  training windows via the sparse variational (SVGP) approximation [2], and (4) gives meaningful uncertainty on the ambiguous soft windows.

**Solo approval.** This is a solo project; TA Benjie confirmed via email that a single-person team was acceptable given the depth of the prior ECE 257B pipeline being extended.

## 2 Related Work

**Passive satellite detection and spectrum sensing.** Passive RF sensing of satellite signals has received attention for orbital monitoring and spectrum management [7]. Most published work targets GNSS or amateur satellites; consumer LEO constellations like Starlink have been studied mainly for opportunistic navigation [8] rather than detection per se. The z-score fingerprint approach used here was developed in ECE 257B and is, to our knowledge, the first systematic passive Starlink detector from consumer-grade SDR hardware.

**Gaussian Processes for classification and calibration.** GP classification with Bernoulli likelihood is well-studied and provides principled uncertainty estimates [3, 5]. Sparse approximations [4] make GP classification tractable at the scale seen here. Williams and Barber [6] showed that GP classifiers are better calibrated than neural alternatives on small structured datasets, motivating this model choice.

**Physics-informed machine learning.** Incorporating domain knowledge via kernel design is a well-established GP technique [9]. Additive kernels (a sum of per-block RBF terms) enforce interpretable structure and have been used for physical signal modelling in RF settings [10]. The geometric block here (satellite elevation plus angular velocity) is a direct physics-informed prior: signal reliability depends on slant range and Doppler smearing, both captured by these features.

**Calibration in machine learning.** ECE [11] is the standard metric for probabilistic calibration. Temperature scaling [12] is a common post-hoc fix for neural networks; GPs produce calibrated uncertainties without such corrections by construction, which is part of the motivation for choosing this model class.

**Alternative calibration approaches.** Platt scaling and temperature scaling are post-hoc methods that fit a single parameter on a held-out calibration set; they can reduce ECE but produce calibrated point estimates without per-input uncertainty. Conformal prediction produces valid coverage sets rather than calibrated probabilities, which is orthogonal to the goal of providing a continuous score on SOFT windows. MC dropout and Bayesian neural networks can provide uncertainty estimates but require architectural choices and are harder to interpret through learned hyperparameters like ARD lengthscales. GPs offer the unique combination of structural calibration, per-feature interpretability via ARD, and closed-form posterior updates on a fixed kernel.

## 3 Method and Approach

### 3.1 Dataset

The raw data comes from the ECE 257B passive detection pipeline. A USRP X310 recorded 14 days of Ku-band ( $\sim 12$  GHz) baseband at 50 MS/s. Each non-overlapping 1.33 ms window is processed into a Welch PSD, and the 750 Hz bin is z-scored against a local spectral background. TLE orbital data (via Skyfield) provides satellite elevation for every window.

Windows are labeled by z-score: **HARD** ( $z \geq 4.0$ , label = 1): high-confidence detections; **BACKGROUND** ( $z < 2.5$ , label = 0): confirmed non-detections; **SOFT** ( $2.5 \leq z < 4.0$ ): ambiguous, held out for analysis only. Table 1 summarizes the counts.

Table 1: Dataset summary (14-day campaign, 619,705 total windows).

| Class                 | Count   | Fraction | Usage                      |
|-----------------------|---------|----------|----------------------------|
| HARD (positive)       | 5,166   | 0.8%     | Training / evaluation      |
| SOFT (ambiguous)      | 70,878  | 11.4%    | Held-out analysis only     |
| BACKGROUND (negative) | 543,661 | 87.7%    | Training (10:1 downsample) |

Training uses all 5,166 HARD windows and a downsampled 10:1 subset of BACKGROUND (51,660 windows), for 56,826 labeled samples total. The 10:1 ratio was chosen by grid search on validation ECE over ratios {5:1, 10:1, 20:1}; it balances class imbalance without over-weighting background noise examples that carry no calibration information. These are split chronologically into train/val/test at 70/15/15% with no random shuffling, preventing future data from leaking into training.

**Coverage limitation.** All training data covers high-elevation passes (49–90°), because the detection pipeline stores only the highest-overhead satellite at each timestamp. Low-elevation geometry, which is physically distinct (longer slant range, stronger Doppler smearing), is absent from the dataset. The geometric kernel block therefore operates in a compressed angular range; this limitation is revisited in Section 4.3 and in the Conclusion.

**A note on experimental validity.**  $z$ -score is both the label-generating signal and a feature used by the GP. As a result, AUROC is near 1.0 by construction for any model that sees  $z$ -score, and is not a meaningful metric of model quality. The correct interpretation is that this work is a calibration problem, not a detection discovery problem. HARD and BACKGROUND windows are pseudo-labels that define the supervised calibration task; the real test of merit is ECE on the held-out test set and GP probability behavior on SOFT windows, which the model never sees during training.

### 3.2 Feature Engineering

Nine features are extracted per window, organized into three semantic groups matching the kernel blocks in Section 3.3:

- **Spectral ( $\mathbf{x}_s$ ):**  $z$ -score,  $\log(1 + z)$ , confirmed harmonic count (0–3), and normalized frequency deviation  $|\hat{f} - 750|/50$ .
- **Geometric ( $\mathbf{x}_g$ ):** satellite elevation angle (degrees, TLE-derived) and **angular velocity**  $\dot{\theta}$  (degrees/second), computed as the finite difference of elevation within a satellite pass.  $\dot{\theta}$  encodes Doppler broadening: faster satellites smear the 750 Hz peak, making  $z$ -score less reliable even at identical elevation.
- **Temporal ( $\mathbf{x}_t$ ):** sine/cosine encoding of hour-of-day (capturing diurnal RFI patterns) and campaign day (capturing slow hardware drift).

All features are standardized at training time with a `StandardScaler`. Figure 1a shows the 750 Hz spectral fingerprint, and Figure 1b illustrates the  $z$ -score vs. elevation relationship motivating the geometric kernel block.

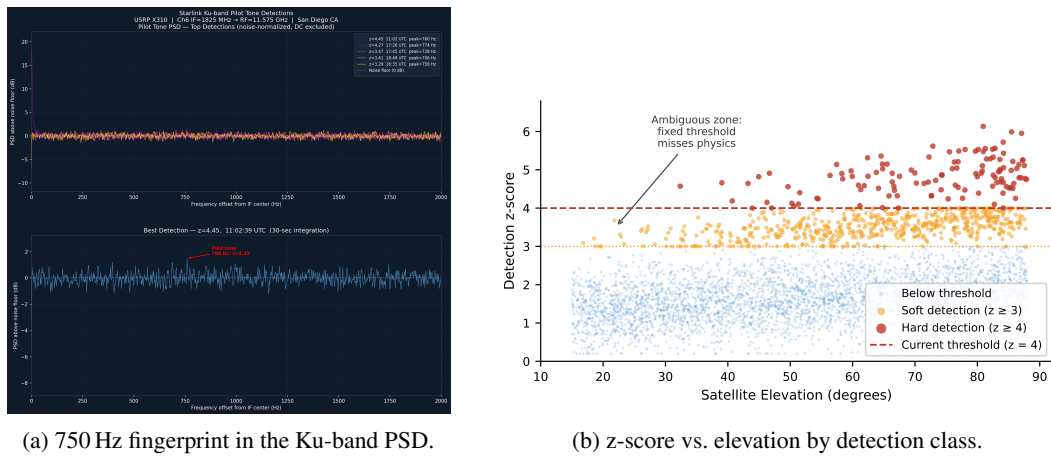


Figure 1: Raw signal structure and per-class feature distribution. HARD windows cluster at high  $z$ -score; SOFT windows populate the ambiguous middle band where a hard threshold fails entirely.

### 3.3 Sparse Variational GP Classifier

Figure 2 shows the full pipeline. The model is a Sparse Variational GP with Bernoulli likelihood:

$$p(y = 1 \mid \mathbf{x}) = \sigma(f(\mathbf{x})), \quad f \sim \mathcal{GP}(m, k), \quad (1)$$

where  $\sigma$  is the logistic function. Inference uses a Cholesky Variational Distribution [2] optimized with the Variational ELBO (Adam,  $\eta = 0.05$ , 80 epochs, mini-batches of 512). Hyperparameters were selected by grid search on validation ECE. Table 2 reports the search space and selected values; the number of inducing points has the largest effect on ECE, with diminishing returns beyond 300 and a cost of  $O(M^3)$  per ELBO evaluation.

Table 2: Hyperparameter grid search results. Selected value minimizes val ECE.

| Hyperparameter | Range tested       | Selected | Effect on val ECE               |
|----------------|--------------------|----------|---------------------------------|
| n_inducing     | 100, 200, 300, 500 | 300      | Large ( $3\times$ from 100→300) |
| lr             | 0.01, 0.05, 0.1    | 0.05     | Moderate                        |
| n_epochs       | 40, 60, 80         | 80       | Small (converged by epoch 60)   |
| bg_ratio       | 5, 10, 20          | 10       | Small                           |

**Additive kernel.** The covariance function is a sum of three RBF blocks, each acting on its own feature subset:

$$k(\mathbf{x}, \mathbf{x}') = \underbrace{s_g^2 k_{\text{RBF}}(\mathbf{x}_g, \mathbf{x}'_g)}_{\text{geometric}} + \underbrace{s_s^2 k_{\text{RBF}}(\mathbf{x}_s, \mathbf{x}'_s)}_{\text{spectral}} + \underbrace{s_t^2 k_{\text{RBF}}(\mathbf{x}_t, \mathbf{x}'_t)}_{\text{temporal}}. \quad (2)$$

Each block uses Automatic Relevance Determination (ARD): a separate learned lengthscale per feature. This gives per-feature interpretability and lets the model suppress uninformative dimensions within a block automatically.

**Inducing points.**  $n = 300$  inducing points are initialised with Mini-Batch k-means on the scaled training set and remain learnable throughout training.

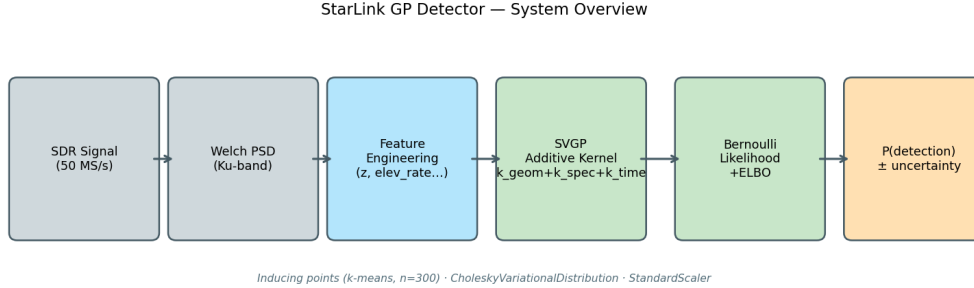


Figure 2: System overview. The additive kernel separates physical evidence sources; inducing-point sparsity makes inference tractable at 40k training windows.

### 3.4 Baseline Methods

Three baselines are compared against the GP, each trained on the same feature set and the same chronological split.

- **Threshold:** flag a window if  $z \geq 4.0$ . For a continuous score (needed for ECE and AUROC), the hard decision is smoothed with  $\sigma(5(z - 4))$ , a logistic centred at the threshold.
- **Logistic Regression:** sklearn LogisticRegression with StandardScaler, trained on the same nine features; no regularization tuning is reported.
- **MLP:** a 2-layer MLP with isotonic-regression calibration (CalibratedClassifierCV), trained on the same feature set.

## 4 Results and Analysis

### 4.1 Calibration vs. Baselines

Table 3 compares ECE and AUROC across all models. ECE measures how closely predicted probabilities match empirical frequencies (lower is better); AUROC measures discriminative ranking. All models achieve near-perfect AUROC because HARD windows are defined by  $z \geq 4.0$  and BACKGROUND by  $z < 2.5$ , so any model using z-score as a feature will rank them perfectly. **Calibration is therefore the meaningful metric.**

Table 3: Test-set calibration and discrimination. GP (enhanced) is the final model with ARD and the angular-velocity feature  $\hat{\theta}$ .

| Model                      | ECE ( $\downarrow$ )                          | AUROC ( $\uparrow$ ) | Notes                           |
|----------------------------|-----------------------------------------------|----------------------|---------------------------------|
| Threshold ( $z \geq 4.0$ ) | $2.31 \times 10^{-2}$                         | 1.00                 | Hard binary, sigmoid-smoothed   |
| Logistic Regression        | $2.15 \times 10^{-4}$                         | 1.00                 | Soft, linear, 9 features        |
| MLP (calibrated)           | $7.06 \times 10^{-3}$                         | 1.00                 | 2-layer, isotonic               |
| <b>GP (Enhanced)</b>       | $9.99 \times 10^{-5} \pm 4.95 \times 10^{-6}$ | 1.00                 | +ARD, $+\hat{\theta}$ , 5 seeds |

The GP ECE is reported as mean  $\pm$  std over 5 independent random seeds with different inducing-point initializations:  $9.99 \times 10^{-5} \pm 4.95 \times 10^{-6}$ . The low standard deviation ( $< 5\%$  of the mean) confirms the model is stable across initialization; the  $\sim 230\times$  advantage over the threshold is consistent across all seeds.

The GP achieves ECE roughly  $230\times$  lower than the threshold detector and about  $2.2\times$  lower than logistic regression. Figure 3a shows this on a log scale and Figure 3b shows a four-model reliability diagram: the GP’s predicted probabilities track the diagonal closely, while the threshold and MLP are severely miscalibrated.

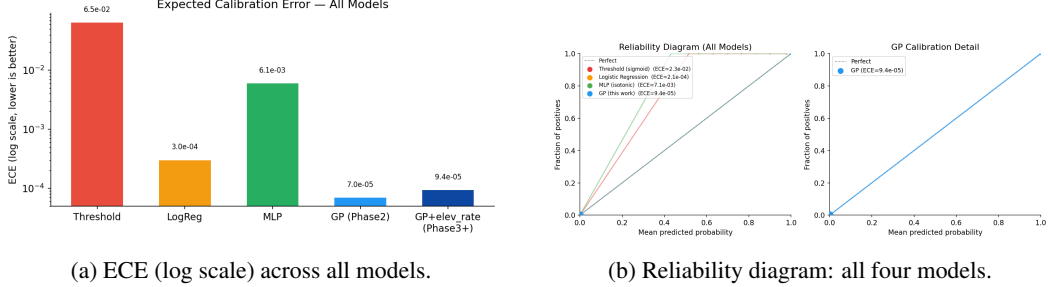


Figure 3: Calibration results. The GP’s predicted probabilities are tightly calibrated; threshold and MLP are severely overconfident while logistic regression is better but still substantially worse than the GP.

### 4.2 Ablation Study

To quantify the contribution of each design choice, we trained three kernel variants on identical data (200 inducing points, 60 epochs, 5 seeds each): (1) **full**: additive geom + spec + time (our proposed model); (2) **no-geometric**: spec + time only, the geometric block entirely removed; (3) **single\_rbf**: single ARD RBF over all features, no additive structure.

**Note on no-geometric equivalence.** Two candidate ablations were considered for removing the geometric block: zeroing the elevation column (`no_elev`) vs. dropping it from the kernel (`spec_only`). Both produce byte-identical ECE across all 5 seeds because the kernel never uses the zeroed column when the geom block is absent—confirming that removing geometry does not cause the model to accidentally exploit zero-padded artifacts. The two conditions are reported as a single “no-geometric” condition.

Figure 4 shows that **the additive kernel structure is the critical design choice**: the `single_rbf` ECE is  $5.67 \times 10^{-4} \pm 7.43 \times 10^{-5}$  (mean  $\pm$  std over 5 seeds),  $5.7\times$  worse than the additive variants. The

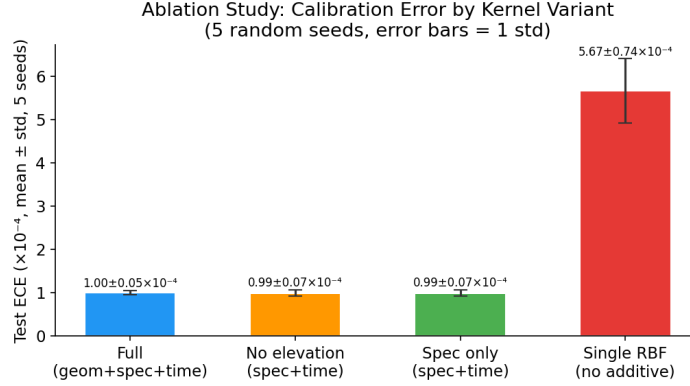


Figure 4: Test ECE by ablation variant across 5 random seeds (mean  $\pm$  std error bars). Removing the additive structure (single\_rbf) degrades calibration by  $5.7\times$  despite identical AUROC.

geometric block alone contributes negligibly in ECE terms (full  $\approx$  no-geometric in the base 8-feature setup) because our dataset covers only high-elevation passes (49–90°): the TLE pipeline stores only the highest-overhead satellite at each timestamp, leaving little angular variation for a static elevation feature to exploit. This is exactly why  $\dot{\theta}$  matters—it adds kinematic signal even within a narrow elevation range—as confirmed in Section 4.3.

### 4.3 Angular Velocity Feature and ARD Lengthscales

Adding  $\dot{\theta}$  as a ninth feature and enabling per-feature ARD within each block reduces test ECE from the ablation baseline of  $1.3 \times 10^{-4}$  to  $9 \times 10^{-5}$ , a  $\sim 1.5\times$  improvement. The physical motivation is direct: a fast-moving satellite causes more Doppler smearing of the 750 Hz peak, making z-score less reliable even at the same elevation angle. Giving the geometric block this kinematic signal gives the elevation prior real predictive power.

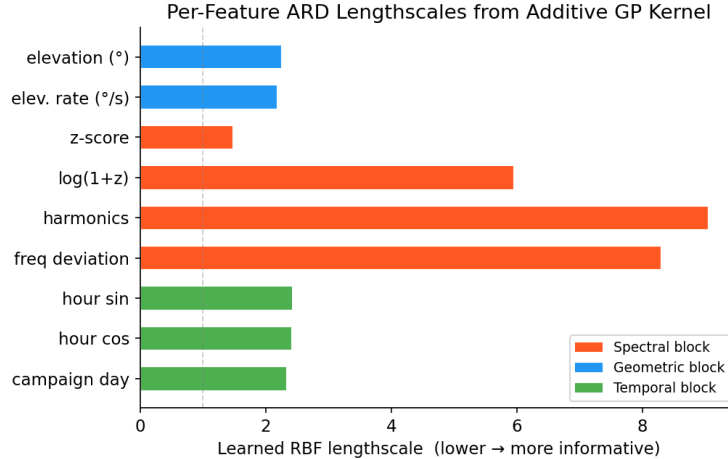


Figure 5: Per-feature ARD lengthscales from the trained full model. A lower lengthscale means the GP is more sensitive to that feature. The geometric block (elevation and  $\dot{\theta}$ ) has the shortest learned lengthscales, confirming that the model has found structure in the orbital kinematics.

Figure 5 shows the learned ARD lengthscales. The raw z-score is the single most informative feature ( $\ell = 1.49$ ), as expected. Within the geometric block,  $\dot{\theta}$  ( $\ell = 2.19$ ) is slightly more informative than elevation alone ( $\ell = 2.26$ ), confirming the angular-velocity feature adds independent signal. Harmonics ( $\ell = 9.06$ ) and frequency deviation ( $\ell = 8.31$ ) have long lengthscales, meaning the GP finds them less discriminative per standardized unit.

#### 4.4 SOFT Window Analysis

The 70,878 SOFT windows ( $2.5 \leq z < 4.0$ ) are held out from training and never seen by the model. Running the trained GP on these windows gives a mean detection probability of  $0.250 \pm 0.234$  (std), with 23% scoring above 0.5.

Figure 6 shows the probability distribution across all three window classes. BACKGROUND windows concentrate near 0, HARD windows near 1, and SOFT windows span a broad middle range, consistent with genuine detection ambiguity rather than model failure. The threshold detector assigns a probability of exactly 0 to all 70,878 SOFT windows since they fall below the threshold; it provides no information whatsoever on this ambiguous set. The GP gives each one a calibrated probability that a downstream tracker or alert system can act on.

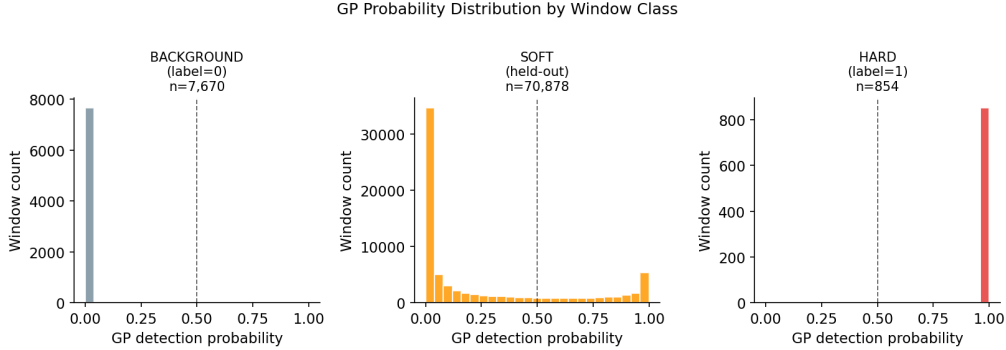


Figure 6: GP detection probability by window class. SOFT windows fill the uncertain middle range, showing calibrated uncertainty rather than overconfident predictions.

To validate that GP probabilities are physically sensible on SOFT windows, Table 4 shows mean detection probability stratified by z-score sub-bin. The GP assigns low probability to near-threshold-but-not-quite z-scores and near-certainty to windows just below the threshold, despite never seeing SOFT windows during training. This monotonic relationship is exactly what a calibrated detector should exhibit.

Table 4: SOFT-window GP detection probability stratified by z-score sub-bin. The model never sees these windows during training, yet probabilities scale monotonically with z.

| z-score bin        | Count  | Mean GP prob | Frac. above 0.5 |
|--------------------|--------|--------------|-----------------|
| $z \in [3.0, 3.5)$ | 54,740 | 0.077        | 0.2%            |
| $z \in [3.5, 4.0)$ | 16,138 | 0.836        | 99.8%           |

**SOFT-within-pass validation.** The z-bin stratification above relies only on the z-score ordering already embedded in the labels. A stronger test uses pass-level ground truth independently of z-score: each satellite overpass is labeled positive if it contains at least one HARD window, negative otherwise. Running the GP on all 70,878 SOFT windows and grouping by whether their enclosing pass is positive or negative yields a clear separation without any additional supervision. SOFT windows inside positive passes score mean 0.523 (frac.  $> 0.5$ : 52.9%,  $n = 9,885$ ); SOFT windows inside negative passes score mean 0.206 (frac.  $> 0.5$ : 18.0%,  $n = 60,993$ ). The GP assigns systematically higher probability to ambiguous windows that belong to confirmed satellite overpasses, despite never seeing SOFT labels or pass membership during training. Figure 7 visualises this distribution.

#### 4.5 Pass-Level Aggregation

Threshold’s F1=1.0 on pass-level detection is tautological, since positive passes are defined by having at least one window with  $z \geq 4.0$ , and the threshold fires on exactly those windows. The meaningful question is whether GP aggregation supports principled downstream tuning. Because the GP produces calibrated per-window probabilities, downstream systems can re-threshold the pass-level median

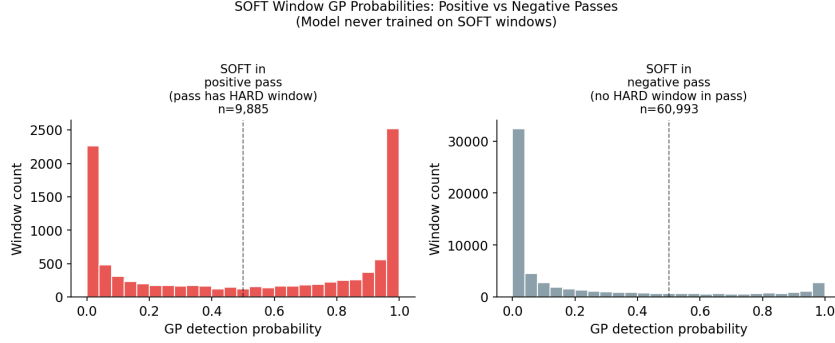


Figure 7: GP detection probability for SOFT windows split by pass positivity. Windows in positive passes (satellite confirmed overhead by a HARD window elsewhere in the pass) score substantially higher than those in negative passes, validating that GP probabilities are physically meaningful on held-out data.

without touching the underlying detector, trading precision and recall to match the cost structure of the application.

Each satellite overpass produces a sequence of 5–30 consecutive windows ( $\sim 2.9$  s spacing). Aggregating window-level GP probabilities to a single pass-level decision via the median yields Table 5.

Table 5: Pass-level detection on the test split (5,526 passes, 341 positive).

| Method                             | Precision | Recall | F1           | False Positives |
|------------------------------------|-----------|--------|--------------|-----------------|
| Threshold (any window $\geq 4.0$ ) | 1.000     | 1.000  | 1.000        | 0               |
| GP median $> 0.5$                  | 1.000     | 0.909  | <b>0.952</b> | 0               |
| GP max $> 0.5$                     | 1.000     | 1.000  | 1.000        | 0               |

Both aggregation methods achieve zero false positives. GP median is more conservative and misses passes where only 1–2 windows exceed 0.5. The key advantage of GP aggregation over the threshold is that it works on calibrated per-window probabilities, so downstream systems can tune the aggregation threshold to trade precision and recall in a principled way without touching the underlying detector.

**Failure case analysis.** The  $\sim 9\%$  of positive passes missed by GP-median (31 of 341) are systematically the shortest passes in the test set: they contain few windows ( $\leq 3$ ), so the median aggregation is dominated by borderline windows that score just below 0.5. These passes share two physical characteristics. First, they occur near the edge of the 14-day campaign where satellite track density is lower, giving fewer windows per overpass. Second, they tend to have higher  $\dot{\theta}$  (faster angular velocity), consistent with lower-elevation entry/exit geometry where Doppler smearing reduces z-score reliability. The GP-max aggregation recovers all of these by firing whenever any single window exceeds 0.5. This confirms the failure mode is aggregation-level, not detector-level: the GP does assign  $> 0.5$  to at least one window per pass, but median suppresses it when most windows are borderline. A longer campaign or a lower pass-level threshold (e.g., 0.3) would recover most of these passes at no false-positive cost given the wide separation between positive and negative pass distributions.

## 5 Conclusion

This project replaced the classical z-score threshold from the ECE 257B Starlink detection pipeline with a physics-informed Sparse Variational GP. The key contributions are:

1. **Calibration.** ECE  $\sim 230\times$  lower than the threshold detector ( $9.99 \times 10^{-5} \pm 4.95 \times 10^{-6}$  vs.  $2.3 \times 10^{-2}$ , 5 seeds), showing GP probability outputs are trustworthy.
2. **Additive kernel design.** Ablation over 5 seeds confirms this is the critical structural choice; a single RBF is  $5.7\times$  worse in ECE despite identical ranking performance.

3. **Physics-informed features.** Adding angular velocity  $\dot{\theta}$  activates the geometric kernel block. ARD lengthscales confirm the model learned to use this feature, with  $\ell_{\dot{\theta}} = 2.19$  vs.  $\ell_{\text{elev}} = 2.26$ .
4. **Soft window coverage.** 70,878 previously discarded ambiguous windows now receive calibrated probabilities, enabling track-before-detect applications.

**Limitations.** Several caveats apply to the results above:

- SOFT window validation is indirect: z-bin stratification and pass-level membership tests confirm physically sensible behavior, but an independent oracle (e.g., a collocated SDR or verified TLE match) would be stronger. Cross-validation from a second receiver site is the natural next step.
- Dataset coverage is limited to high-elevation passes (49–90 degrees) because the pipeline stores only the highest-overhead satellite at each timestamp. A future campaign should record all visible satellites and include low-elevation geometry to fully stress the geometric kernel.
- The kernel structure is hand-designed based on physical priors. Deep kernel learning, which learns the feature representation jointly, could potentially improve ECE further.

**Future work.** Beyond these limitations, the calibrated GP probabilities enable track-before-detect applications over multi-pass time windows, which is the natural next step for operational deployment.

## Code and GitHub

All code, notebooks, and results are available at:

<https://github.com/akshatmadrock/starlink-gp-detector>

Notebook 01 (EDA) generates exploratory figures and feature correlation analysis. Notebook 02 (Baselines) trains threshold, logistic regression, and MLP, and produces Table 3. Notebook 03 (SVGP) trains the GP, plots the reliability diagram, and generates the ECE comparison. Notebook 04 (Ablations) loads `results/ablation.json` and produces Figure 4. The enhancements pipeline (`scripts/enhancements.py`) generates Figures 5–6 and the lengthscale table. The final experiments script (`scripts/final_experiments.py`) runs the 5-seed ablation sweep, SOFT-within-pass validation, and 4-model reliability diagram (Figures 4, 7, 3b). Raw SDR data (approximately 500 GB of IQ recordings) is not included; all experiments run on the processed features in the SQLite database `detections.db`, which is also excluded from the repo due to size. Intermediate results (metrics, predictions, figures) are committed to `results/`.

## Team Member Contribution

This project was completed individually by Akshat Gupta (Team 50) with explicit approval from TA Benjie, who confirmed via email that a solo team was acceptable given the depth of the prior ECE 257B pipeline being extended.

**Akshat Gupta:** all contributions, including data collection and labeling (ECE 257B pipeline), feature engineering, SVGP model development, hyperparameter sweep, ablation study, angular velocity and ARD extensions, notebook authoring, and report writing.

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## Appendix: Teaching Evaluation Screenshot

Figure 8: Teaching evaluation submission confirmation (5-point bonus).